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1. Executive Summary

This report investigates ethical, statistical, and machine-learning approaches to insurance data analytics. It begins by outlining the principles that govern responsible algorithmic use, including privacy protection, fairness, transparency, human oversight, and the promotion of societal solidarity. These principles ensure that data-driven systems remain trustworthy and compliant with regulatory expectations.

Building upon this foundation, Section [5](#) analyses travel-insurance loss data. Data quality issues (e.g., outliers, missing values) are addressed, and demographic rating structures are explored. Loss severity is stabilized using a Box–Cox transformation, where the estimated parameter is $\hat{\lambda} \approx 0.02$, indicating a near-log transformation. Age-banding is subsequently applied, and mean claim severity is compared across genders, with male claimants exhibiting marginally higher losses.

Section [6](#) applies principal component analysis to home-insurance severity predictors. Approximately 88.6% of variation is captured by the first three principal components, supporting dimensionality reduction. Component loadings demonstrate that property age and distance contribute most to early principal directions. Principal component regression, while exhibiting low explanatory power, is used to predict cost for a hypothetical new policy, yielding approximately 178 (thousand monetary units).

Section [7](#) models motor-insurance claim frequency using a Poisson neural network with embedding layers for categorical factors and a log-exposure offset. Deviance loss is used as the evaluation metric; training and testing losses of 0.3117 and 0.3069 show limited overfitting and reasonable predictive stability. The framework retains actuarial familiarity, while leveraging deep-learning flexibility.

Overall, the report demonstrates an end-to-end analytical process integrating algorithmic-ethics principles with statistical modelling and neural-network methods. Findings indicate that while classical techniques remain essential for interpretability and governance, deep neural networks provide enhanced representational power with competitive predictive performance. Ethical considerations remain central throughout, ensuring that analytical outputs are responsibly

deployed in insurance contexts.

2. Introduction

The rapid adoption of data-driven methods in insurance — including machine learning, representation learning, and automated decision support — has accelerated the demand for responsible AI governance. Insurers now face dual expectations: to realize the benefits of predictive modelling, while also safeguarding societal values such as privacy, fairness, and solidarity.

Section [4](#) of this report addresses these expectations by outlining the key ethical and regulatory principles that govern algorithmic use. These include privacy preservation, fairness and non-discrimination, transparency and explicability, human-in-the-loop oversight, and solidarity and inclusion. Within the insurance sector, such principles are embedded in international and regional regulatory frameworks, notably the EU Artificial Intelligence Act (European Commission, 2024) and associated data-protection legislation such as the GDPR. Their collective purpose is to ensure that algorithmic systems remain accountable, interpretable, and trustworthy, particularly when applied to pricing, reserving, underwriting, and fraud detection.

The remaining parts of this report operationalize these principles through quantitative workflows that combine classical actuarial methods with modern machine-learning techniques. Section [5](#) analyses travel-insurance loss data through structured data cleaning, outlier detection, and demographic segmentation. A Box–Cox transformation (Box and Cox, 1964) is applied to stabilize highly skewed claim amounts, improving their suitability for downstream modelling. Section [6](#) investigates home-insurance costs using principal component analysis (PCA) (James et al., 2021) to reduce multicollinearity and uncover latent structure among rating factors such as property value, property age, distance, and policy duration. A principal-component regression is then fitted, and prediction is demonstrated for a hypothetical new policy. Section [7](#) focuses on motor-insurance claim frequency, modelled via a Poisson deep neural network (DNN) (Wüthrich and Merz, 2021; Goodfellow et al., 2016). The network incorporates continuous predictors and embedded categorical representations, together with a log-exposure offset to preserve actuarial interpretability. Model performance is evaluated using deviance loss, consistent with generalized linear modelling (GLM) methodology (Nelder and Wedderburn, 1972).

Together, these components form a modern actuarial analytics pipeline:

- (1) beginning with ethical and regulatory foundations;
- (2) progressing to structured data preparation;
- (3) performing dimensionality reduction and exploratory analysis; and
- (4) advancing to neural-network-based prediction with explicit actuarial controls.

The report emphasizes that ethical governance and statistical performance are complementary, not competing, priorities. Sound model design, transparency, and fairness must underpin every quantitative procedure—assuring that advanced analytical methods contribute to both technical robustness and social responsibility within the insurance industry.

3. Data and Methods

3.1 Data Sources

Three datasets were analyzed:

1. Travel insurance dataset (travel_insurance.xlsx):
Contains claimant-level information (e.g., age, gender, attorney involvement, accident type, claim severity).
Used in Section [5](#) – Data Wrangling & Transformation.
2. Property insurance dataset (property.xlsx):
Contains property-level information (e.g., property value, property age, distance, and claim outcomes).
Used in Section [6](#) – Principal Component Analysis (PCA) & PC Regression.
3. Motor insurance dataset (freMTPL2freq.csv):
Contains policy-level risk attributes (e.g., driver age, vehicle characteristics, exposure, claim counts).
Used in Section [7](#) – Poisson Neural Network Modelling.

All datasets were provided in spreadsheet or CSV format and imported into R.

3.2 Software Environment

All computation was performed in R version 4.5.1 (R Core Team, 2023) within RStudio. To ensure reproducibility, a global random seed was set (set.seed(100)). Major packages used include tidyverse, dplyr, readxl, MASS, and ggplot2 for data wrangling and visualization; stats for PCA and

regression analysis; and keras3 with reticulate for neural-network modelling. A complete list of packages and dependencies is provided in Appendix A, along with full commented R code contained in *appendix.txt*.

4. Regulations and Data Science Ethics

4.1 Introduction

The rapid adoption of machine learning in insurance has transformed underwriting, pricing, claims assessment, and fraud detection, improving efficiency and enabling real-time data processing. However, these algorithmic systems raise significant ethical concerns regarding privacy, fairness, and societal solidarity, which must be addressed to protect policyholders' rights and maintain trust in insurance markets (European Commission, [2024](#)). Given their influence on access to financial services and individual economic welfare, effective ethical governance has become essential (World Economic Forum, [2021](#)). This section outlines the key ethical principles of privacy, fairness, and solidarity, and summarizes practical strategies insurers can adopt to meet regulatory and ethical standards.

4.2 Key Ethical Principles

Privacy

Privacy entails individuals' control over how their personal data are collected, processed, and retained (European Commission, [2024](#)). In insurance, this includes demographic, financial, medical, and behavioral data. Ethical use demands adherence to purpose limitation and secure handling through encryption, anonymization, and restricted access (GDPR, [2016](#)). Privacy-preserving techniques such as differential privacy (Dwork and Roth, [2014](#)) and federated learning (Li et al., 2020) enable statistical analysis while minimizing personal data exposure. Transparent disclosure of data usage supports accountability and public trust (Richards and King, [2014](#)).

Fairness

Fairness requires that algorithmic decisions do not disproportionately disadvantage protected groups (Barocas and Selbst, [2016](#)). In insurance, biased training data or proxy variables can embed structural inequities (O'Neil, [2016](#)). Ensuring fairness involves bias detection, counterfactual testing, fairness-constrained optimization, and independent audits (Raji et al., [2020](#)). Transparency and explainability—through interpretable models or post-hoc tools such as LIME and SHAP—allow

regulators and policyholders to contest outcomes (Wachter et al., [2017](#)). Within actuarial practice, fairness must coexist with risk-based pricing so that individual assessment does not reinforce societal inequality (Schwarcz, [2021](#)).

Solidarity

Solidarity underpins insurance by pooling risk across populations (Taylor, [2019](#)). Increasingly granular data and personalized pricing threaten this principle by reducing mutualization and affordability for higher-risk groups (Schwarcz, [2021](#)). Ethical practice therefore balances actuarial precision with inclusiveness. Measures such as limiting excessive segmentation, applying cross-subsidies, or maintaining minimum coverage standards preserve accessibility (Baker, [2020](#)). Solidarity also guides product design, encouraging community-rated or social insurance structures that protect vulnerable groups.

4.3 Strategies and Practices in Insurance

Protecting Privacy

Insurers can safeguard privacy through data minimization, encryption, secure storage, and restricted access (GDPR, [2016](#)). De-identification and data aggregation reduce re-identification risks, while differential privacy enables statistical analysis without exposing individual records (Dwork and Roth, [2014](#)). Transparent consent management ensures that policyholders understand and control how their data are used (Richards and King, [2014](#)).

Ensuring Fairness in Decision-Making

Fairness can be upheld through regular bias audits, sensitivity testing, and fairness monitoring (Raji et al., [2020](#)). Many insurers now establish ethics or governance committees to review model outputs and ensure compliance with regulatory and ethical standards (World Economic Forum, [2021](#)). Explainability techniques—such as LIME and SHAP—enhance transparency, enabling both regulators and customers to interpret model behavior (Wachter et al., [2017](#)).

Fostering Solidarity

To preserve solidarity, insurers should limit excessive segmentation by restricting highly granular risk factors or capping rate differentials (Schwarcz, [2021](#)). Regulatory measures such as minimum benefit standards or risk-sharing pools can maintain affordability (Baker, [2020](#)). Solidarity can also be reinforced through inclusive product design, such as community-rated or social insurance models that protect vulnerable groups (Taylor, [2019](#)).

4.4 Conclusion

Algorithmic systems can enhance efficiency, pricing accuracy, and claims management, but they must operate within ethical boundaries. Protecting privacy, ensuring fairness, and maintaining solidarity remain core obligations of insurers. Lawful data practices, transparency, and bias mitigation are essential for responsible innovation. As predictive technologies evolve, robust governance frameworks and ethical oversight will be vital to sustain public trust and uphold the social purpose of insurance (European Commission, [2024](#); GDPR, [2016](#); Schwarcz, [2021](#)).

5. Data Manipulation and Wrangling

5.1 Structure of Data Set

The *travel_insurance* dataset consists of 1,340 observations and six variables (caseld, attorney, claimantGender, containerAccident_type, claimantAge, and loss). Basic summaries reveal missingness across claimantGender (12), containerAccident_type (48), and claimantAge (189).

5.2 Outlier handling

Reported loss values range from 0.005 to 1,000,000.265 (units in thousands), exhibiting extreme right-skewness with a mean of 3,737 and a median of 2.350. To mitigate the influence of extreme values while maintaining data completeness, outliers were defined using the empirical 0.5th and 99.5th percentiles of loss:

$$q_{0.5} = 0.0300, q_{99.5} = 238.0207.$$

Observations with losses outside this range (13 cases) were removed (set to NA then filtered), along with negative losses. In all, >99% of the dataset remained. This approach provides robust trimming while preserving distributional structure required for Box–Cox transformation.

Formally, if L denotes loss and $(q_{0.5}, q_{99.5})$ the empirical percentiles, the retained sample is:

$$\{i: q_{0.5} \leq L_i \leq q_{99.5}\}, \text{ with } q_{0.5} = 0.0300, q_{99.5} = 238.0207.$$

5.3 Removing Negative Loss Values

Following the identification of extreme loss values in Section [5.2](#), all records corresponding to these outliers were deleted from the dataset. This step ensures that highly atypical observations do not distort subsequent statistical analyses or transformations.

The question further specifies that negative claim amounts should also be removed. In insurance datasets, negative losses occasionally occur due to refunds, claim reversals, or accounting adjustments. Although such transactions may have operational relevance, they do not represent true claim severities and are therefore excluded from this analysis.

By removing both the previously identified outlying losses and all negative values, the dataset is restricted to valid, non-negative claim amounts. This refined dataset accurately reflects realistic claim behavior and provides a reliable basis for subsequent modelling stages, including the Box–Cox transformation and the analysis of claim severity patterns.

5.4 Box–Cox transformation

The Box–Cox transformation was applied to the loss variable to reduce skewness and stabilize variance. Because the transformation requires strictly positive input values, zero losses were replaced by a small constant (0.0001) before estimation to ensure positivity.

Let L_i denote the adjusted positive loss for observation i .

The Box–Cox transformation is defined as

$$L_i^{(\lambda)} = \begin{cases} \frac{(L_i)^\lambda - 1}{\lambda}, & \text{if } \lambda \neq 0, \\ \log L_i, & \text{if } \lambda = 0. \end{cases} \quad (1)$$

The optimal transformation parameter $\hat{\lambda}$ is obtained by maximum likelihood under the model $L_i \sim 1$ (an intercept-only model). Specifically, the estimate maximises the log-likelihood function

$$\log(\lambda) = -\frac{n}{2} \log(\hat{\sigma}_\lambda^2) + (\lambda - 1) \sum_{i=1}^n \log L_i, \quad (2)$$

where $\hat{\sigma}_\lambda^2$ is the residual variance of the transformed variable $L^{(\lambda)}$ for a given λ .

The maximum-likelihood estimate was

$$\hat{\lambda} \approx 0.0202.$$

Because this value is close to zero, the logarithmic transformation

($\lambda = 0$) is nearly optimal. Consequently, a log-scale approximation was adopted in later modelling to achieve approximate normality and homoscedasticity of the transformed loss distribution.

Figure 1 illustrates the log-likelihood profile for the Box–Cox transformation parameter λ . The curve attains its maximum near $\hat{\lambda} \approx 0.02$, with a flat peak around zero, indicating that a logarithmic transformation provides an almost optimal variance-stabilizing effect. The horizontal line marks the 95% confidence region, confirming that values of λ close to zero yield statistically equivalent fits. This visual evidence supports adopting the log transformation for subsequent analyses.

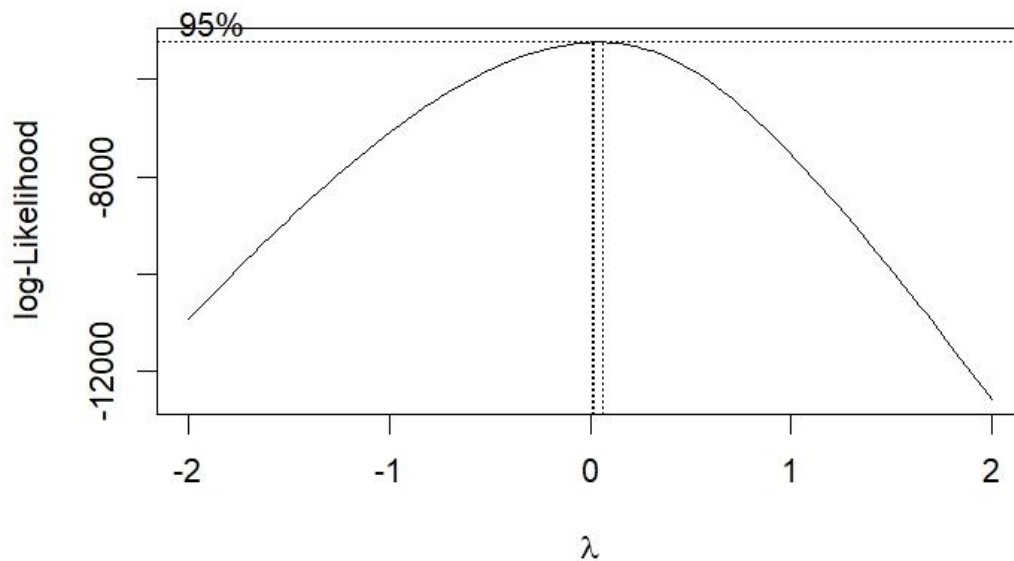


Figure 1. Profile of the Box–Cox log-likelihood function used to estimate the optimal transformation parameter $\hat{\lambda} \approx 0.02$.

5.5 Loss by gender

After cleaning, mean losses by gender are: Male = 5.66, Female = 4.43 (thousands). The difference is descriptive rather than causal and reflects both underlying claim distributions and any remaining data heterogeneity; a formal comparison would require modelling adjustments for age, accident type, and potential selection effects.

5.6 Age banding

To facilitate demographic analysis and enable comparison of claim behavior across different age ranges, a categorical variable was created from the continuous age variable claimantAge.

The age of each claimant was grouped into ordered categories representing distinct life stages that may correspond to different risk profiles:

- ≤ 25 years – young claimants (strictly under 25);
- 26–35 years – early working age;
- 36–42 years – mid-career individuals.
- 43–72 years – mature working or early retirement ages;
- > 72 years – elderly claimants.

Each record was assigned to one of these intervals, forming a new factor variable age group with an ordered structure reflecting the natural progression of age.

The observed counts were:

Age Group (years)	Frequency
≤ 25	442
26 – 35	227
36 – 42	171
43 – 72	279
≥ 73	22

Table 1. *Distribution of claimants by age group.*

The resulting distribution showed that most claimants were between 18 and 72 years old, with the highest frequency in the ≤ 25 category (442 observations) and progressively fewer in older groups. An adjustment (≤ 25) was made to avoid an open lower boundary at 25 and is reported transparently for reproducibility.

6. Principal Component Analysis (PCA) and Linear Regression

6.1 Correlation

A Pearson correlation matrix was constructed using all numeric variables within the property-insurance dataset to explore linear relationships among predictors and the response variable (claimCost). Table [2](#) summarizes the pairwise correlations.

Interpretation

Two notable relationships emerge from the correlation scan:

- Property age vs. property value (correlation = -0.5435):
A moderate negative correlation indicates that older properties tend to be insured for lower values. This pattern is intuitive, as building depreciation, wear, and outdated construction materials typically reduce the replacement cost or market value of older structures.
- Claim count vs. claim cost (correlation = 0.4818):
A moderate positive correlation shows that policies with more claims also incur higher total claim costs. This reflects an expected operational reality — a greater frequency of claims generally results in higher aggregate loss expenditure, even when individual claims may vary in severity.

All other correlations are weak, suggesting limited multicollinearity among most predictors. This confirms the suitability of applying Principal Component Analysis (PCA) in the next stage to capture underlying patterns and reduce dimensional redundancy.

Variable	propertyVal	claimCount	claimCost	propertyAge	distance	duration
propertyVal	1	0.018	0.0098	-0.5435	-0.0049	-0.0017
claimCount	0.018	1	0.4818	-0.0116	-0.0042	-0.0020
claimCost	0.0098	0.4818	1	-0.0000	0.0051	-0.0027
propertyAge	-0.5435	-0.0116	-0.0000	1	0.005	0.001
distance	-0.0049	-0.0042	0.0051	0.005	1	-0.0025
duration	-0.0017	-0.0020	-0.0027	0.001	-0.0025	1

Table 2. Pearson correlation matrix for property-insurance variables.

6.2 PCA and variance retention

Principal Component Analysis (PCA) was performed on the standardized predictors

propertyVal, propertyAge, distance, duration

to reduce dimensionality and identify orthogonal components that explain most of the variability in the dataset. Before conducting PCA, all variables were standardized to ensure comparability, as features measured on different scales can otherwise distort the principal components. The standardized design matrix is denoted by

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}, \text{ where each column has mean 0 and variance 1.}$$

The correlation matrix of the standardized variables is

$$R = \frac{1}{n-1} X^T X. \quad (3)$$

The PCA procedure determines a set of orthonormal eigenvectors

$$v_1, v_2, \dots, v_p \in \mathbb{R}^p$$

and corresponding eigenvalues

$\lambda_1, \lambda_2, \dots, \lambda_p$, which satisfy

$$Rv_j = \lambda_j v_j, j = 1, 2, \dots, p.$$

Each eigenvector defines a principal component direction, and each eigenvalue represents the variance explained by that component.

If s_j denotes the standard deviation of the j^{th} principal component, then $\lambda_j = s_j^2$.

The cumulative proportion of variance explained by the first k components is given by

$$\text{CumVar}(k) = \frac{\sum_{j=1}^k \lambda_j}{\sum_{j=1}^p \lambda_j} = \frac{\sum_{j=1}^k s_j^2}{\sum_{j=1}^p s_j^2}. \quad (4)$$

Results

The computed eigenvalues and cumulative proportions of variance are summarized in Table [3](#).

Principal Component	Cumulative Variance Explained
PC1	0.386
PC2	0.637
PC3	0.886
PC4	1

Table 3. Proportion of total variance explained by successive principal components from the property-insurance PCA.

Thus, the first three principal components together account for approximately 88.6% of the total variation, exceeding the conventional 80% retention threshold. Consequently, the first three components were retained for subsequent modelling.

Interpretation

The rapid increase in cumulative variance from PC1 to PC3 indicates that most of the structural information in the dataset is captured by the first few orthogonal directions. This reduction in dimensionality simplifies the regression task while maintaining most of the data's explanatory power. Since the remaining component contributes minimal additional variance, its exclusion avoids unnecessary noise and potential overfitting in downstream analysis.

6.3 PCA biplot interpretation

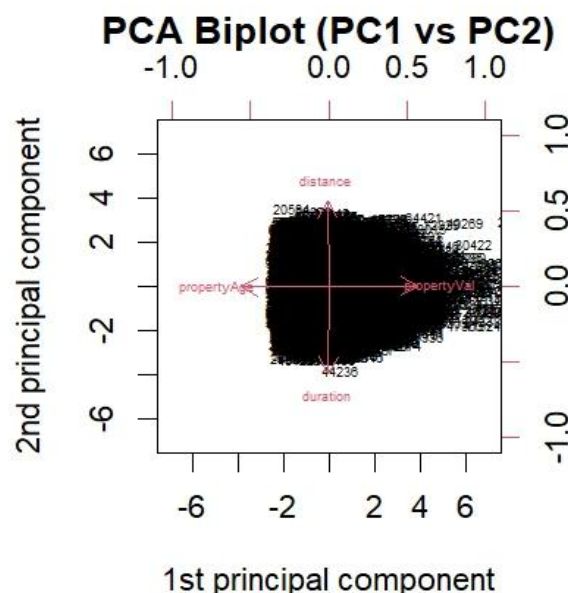


Figure 2. PCA biplot showing the first two principal components (PC1 vs PC2).

PC1 largely contrasts propertyVal against propertyAge:

- Their opposite arrow directions reflect strong negative correlation.
- Distance loads primarily on PC2, creating an orthogonal dimension.
- Duration contributes weakly to PC1–PC2 and is represented with a shorter vector.

The score cloud shows no strong clustering, indicating relatively continuous structure. PCA therefore reveals two interpretable axes:

1. Value–Age gradient (PC1)
2. Distance gradient (PC2)

6.4 Principal-component regression

A regression of claim cost on the first three PCs:

$$\text{claimCost}_i = \beta_0 + \beta_1 Z_{1i} + \beta_2 Z_{2i} + \beta_3 Z_{3i} + \varepsilon_i, \quad (5)$$

Term	Estimate	Std. Error	<i>t</i> value	<i>p</i> value	Significance
(Intercept)	137.27	4.055	33.852	< 0.001	***
Comp. 1	4.714	3.264	1.444	0.149	–
Comp. 2	5.807	4.05	1.434	0.152	–
Comp. 3	1.91	4.06	0.471	0.638	–
Model statistic					Value
Residual standard error					1056
Multiple R^2					0.000064
Adjusted R^2					0.00002
F-statistic (3, 67852 df)					1.455
Overall <i>p</i> -value					0.225

Table 4. Summary of principal component regression on claim cost.

The regression summary indicated that the intercept was statistically significant, but none of the principal components showed strong relationships with claim cost ($p > 0.05$). The model's low coefficient of determination ($R^2 \approx 0.00006$) suggests that, although the principal components capture most of the variance among the predictors, they explain very little of the variation in claim cost. This is a common feature of insurance data, where claim amounts are highly skewed, zero-inflated, and driven by factors beyond the basic property characteristics included here.

The poor model fit reflects a near-zero linear signal between the extracted components and claim cost, likely arising from:

- extreme zero inflation (many zero or very small claim costs), and
- heavy-tailed loss distributions (few but very large claims).

Nevertheless, the Principal Component Regression (PCR) remains useful in this context because:

1. it produces a dimension-reduced representation of correlated predictors,
2. it removes multicollinearity, leading to more stable coefficient estimates, and
3. it quantifies how much of the response variation can be explained by the main orthogonal directions in the feature space.

Overall, while the predictive power of this linear component model is limited, the analysis confirms that PCA effectively summarized the underlying structure of the property attributes. It highlights the typical trade-off between interpretability and predictive accuracy in linear, component-based modelling of insurance data.

6.5 New policy prediction

After fitting the Principal Component Regression (PCR) model in Section 6.4, a prediction was made for a new property-insurance policy using the same set of predictor variables: property value, property age, distance, and duration. The aim was to demonstrate how the PCA transformation and regression model can be applied to unseen data for claim-cost estimation.

The new policy characteristics were:

- Property value = 22
- Property age = 14.5 years
- Distance = 25
- Duration = 10

Since PCA was performed on standardized predictors, the same centering and scaling parameters from the original PCA model were applied to the new data to ensure consistency. Formally, the standardization step computes:

$$x_j^* = \frac{x_j - \bar{x}_j}{s_j}, \quad (6)$$

where

- x_j is the raw input value for predictor j ,
- $\bar{x}_j$ is the mean of that variable in the training dataset, and
- s_j is its standard deviation.

This transformation aligns the new observation with the same scale used during model training, ensuring that it can be projected correctly into the principal component (PC) space.

Next, the projection step converts the scaled inputs into principal component scores:

$$z_{\text{new}} = x_{\text{new}}^* P, \quad (7)$$

where

P is the loading matrix containing the eigenvectors (or weights) corresponding to the retained principal components.

The resulting PC scores (z_1, z_2, z_3) represent the new policy in the reduced three-dimensional space defined by the first three principal components. These scores were then supplied to the previously estimated regression model:

$$\hat{y}_{\text{new}} = \hat{\beta}_0 + \hat{\beta}_1 z_1 + \hat{\beta}_2 z_2 + \hat{\beta}_3 z_3,$$

yielding a predicted claim cost of:

$$\hat{y}_{\text{new}} = 178.00,$$

expressed in the same monetary units as the training data (i.e., approximately 178 thousand).

Interpretation

The result indicates that, based on the property's characteristics and the PCA regression model, the expected claim cost for this new policy is 178.00.

While the prediction accuracy is limited (as noted from the weak regression fit in Section [6.4](#), this example demonstrates the end-to-end PCA regression workflow — from data standardization, projection into component space, and linear prediction — providing a practical framework for extending model insights to new observations.

7. Deep Neural Networks

7.1 Feature Engineering and Model Design (Frequency)

Model motor claim frequency $Y_i = \text{ClaimNb}_i$ with a Poisson mean constrained by exposure. The working dataset (678,013 rows; 12 variables) contains continuous (e.g., DrivAge, BonusMalus, VehPower, VehAge, Density) and categorical attributes (VehBrand, VehGas, Area, Region) plus Exposure. Consistent with the coursework brief:

- Capping: ClaimNb is truncated at 4 and Exposure at time 1 to limit undue influence of extreme values and time.
- Train/test split: a 90/10 split is created with `set.seed(100)` for reproducibility.

Continuous design (min-max to $[-1,1]$)

All continuous predictors—and two categorical fields recoded as integers—are scaled to $[-1,1]$ using

$$x^* = 2 \frac{x - \min(x)}{\max(x) - \min(x)} - 1. \quad (8)$$

VehGas and Area are not embedded but treated as scaled continuous inputs (via their integer codes), reducing model complexity while retaining signal.

Embeddings (categoricals with many levels)

Two higher-cardinality factors are embedded:

- VehBrand (11 levels) $\rightarrow \mathbb{R}^2$,
- Region (22 levels) $\rightarrow \mathbb{R}^2$.

Zero-based integer indices feed learnable embedding layers (dimension $q_{\text{Emb}} = 2$ each). This balances parsimony and flexibility: learned dense vectors capture latent similarity among brands and regions that one-hot encodings would not.

7.2 Model Fitting and Assessment (Poisson Deviance)

The Poisson neural network was trained to model the claim frequency $Y_i = \text{ClaimNb}_i$ using a log-link function and an exposure offset. The model architecture combined the continuous design matrix (scaled to $[-1, 1]$) with the embedded categorical features (VehBrand and Region).

The neural predictor consisted of three fully connected hidden layers with dimensions $q_1 = 32$, $q_2 = 24$, and $q_3 = 16$. Each layer employed the hyperbolic tangent activation function:

$$h^{(\ell)} = \tanh(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)}), \quad (9)$$

where $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ is a smooth, bounded nonlinearity that maps inputs to $[-1, 1]$, providing numerical stability and mitigating the effects of outliers or heterogeneous feature scales.

The network output $\eta_i = f_\theta(x_i)$ represents the log-mean of the Poisson distribution. Incorporating the exposure offset $\log v_i$, the model predicts:

$$\mu_i = \exp(\eta_i + \log v_i) = v_i \exp(\eta_i), \quad (10)$$

so that the expected number of claims remains proportional to exposure, consistent with the classical Generalized Linear Model (GLM) formulation (McCullagh and Nelder, 1989).

The model was trained via gradient descent using the Nadam optimizer for 200 epochs with a batch size of 10 000, minimizing the Poisson deviance loss function.

Poisson Deviance Formulation

For observation i with count y_i and fitted mean $\hat{\mu}_i$, the per-observation deviance is defined as

$$d_i = -2[\ell(y_i; \hat{\mu}_i, \phi) - \ell(y_i; y_i, \phi)] = 2[\log f(y_i | \mu_i = y_i) - \log f(y_i | \mu_i = \hat{\mu}_i)], \quad (11)$$

where $\ell = \log f$ denotes the log-likelihood of the Poisson density and ϕ is the dispersion parameter (fixed to 1 for Poisson).

The mean deviance over a sample $\mathcal{A}$ (training or test) is then

$$\mathcal{L}_{\mathcal{A}} = \frac{2}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} [\log f(y_i | \mu_i = y_i) - \log f(y_i | \mu_i = \hat{\mu}_i)] \quad (12)$$

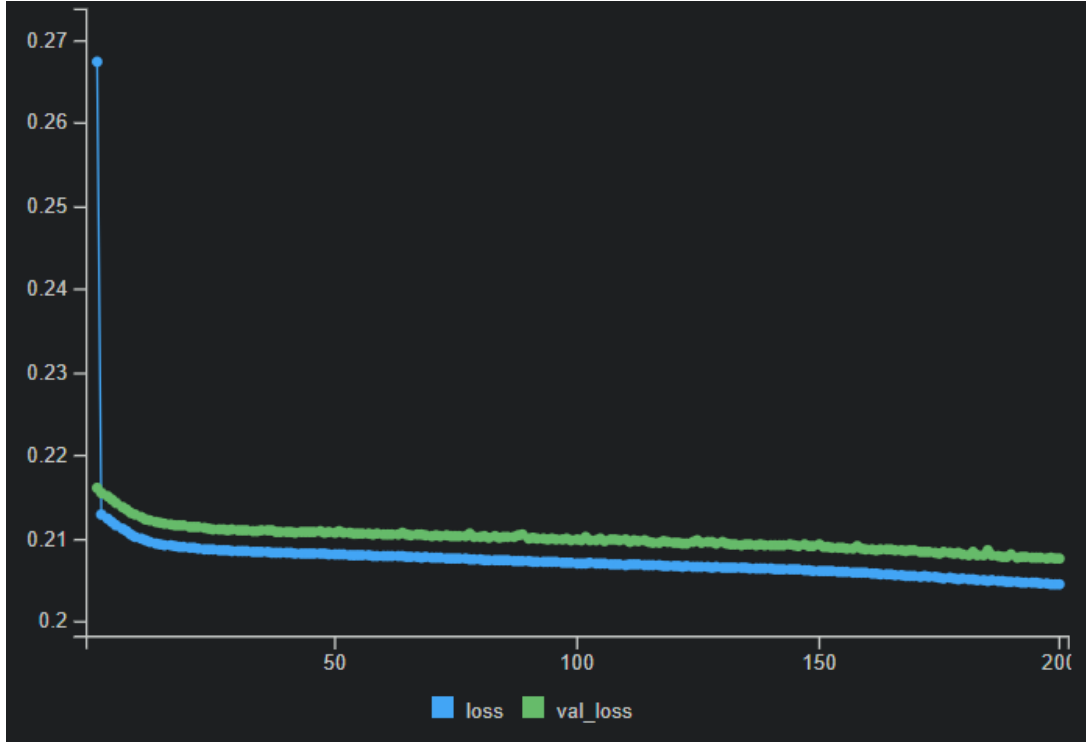


Figure 3. Training and validation Poisson deviance loss across 200 epochs.

Results:

- $\bar{D}_{\text{train}} = 0.3117$
- $\bar{D}_{\text{test}} = 0.3069$

The close similarity between the training and test deviances indicates that the model generalized well and did not overfit. The relatively low deviance values reflect stable learning and proper calibration of expected claim counts. The use of hyperbolic tangent activations ensured smooth nonlinear transformations of scaled features, improving gradient stability and interpretability. Compared with standard linear GLMs, this architecture captured subtle nonlinear relationships among risk factors while maintaining actuarial interpretability through the exposure offset.

7.3 Training a network with Poisson–Inverse Gaussian (P–IG) deviance, $\sigma = 1.2$

Model and likelihood

Model non-negative claim counts $Y \in \{0,1,2, \dots\}$ with mean $\mu > 0$ and fixed dispersion $\sigma > 0$ by the Poisson–Inverse Gaussian distribution, given the pmf

$$\Pr(Y = k \mid \mu, \sigma) = \left(\frac{2\alpha}{\pi}\right)^{1/2} \frac{e^{1/\sigma}}{k!} \left(\frac{\mu}{\alpha\sigma}\right)^k K_{k-\frac{1}{2}}(\alpha), \alpha^2 = \frac{1}{\sigma^2} + \frac{2\mu}{\sigma}, k = 0,1,2, \dots,$$

where $K_\lambda(\cdot)$ is the modified Bessel function of the third kind.

The network with the canonical log link and the exposure offset,

$$\eta_i = f_\theta(\mathbf{x}_i) + \log(\text{Exposure}_i), \mu_i = \exp(\eta_i).$$

For one observation (y, μ) the log-likelihood (up to constants we keep here for exactness) is

$$\ell(y; \mu, \sigma) = \frac{1}{2} \log \left(\frac{2\alpha}{\pi} \right) + \frac{1}{\sigma} - \log(y!) + y(\log \mu - \log(\alpha\sigma)) + \log K_{y-\frac{1}{2}}(\alpha), \alpha = \sqrt{\frac{1}{\sigma^2} + \frac{2\mu}{\sigma}}.$$

Throughout this part we take $\sigma = 1.2$ as fixed, so $\alpha = \alpha(\mu)$ varies only through μ .

Training objective (loss)

Given data $\{(y_i, \mathbf{x}_i, \text{Exposure}_i)\}_{i=1}^n$, the training loss is the average negative log-likelihood

$$\mathcal{L}(\theta) = -\frac{1}{n} \sum_{i=1}^n \ell(y_i; \mu_i(\theta), \sigma), \mu_i(\theta) = \exp(f_\theta(\mathbf{x}_i) + \log \text{Exposure}_i)$$

(with optional regularization added to $\mathcal{L}$ if desired). Minimising $\mathcal{L}(\theta)$ trains the network.

Derivatives for backpropagation

Need $\partial \ell / \partial \eta$ per observation; chain rule then gives gradients w.r.t. θ .

Useful identities

From $\alpha^2 = \sigma^{-2} + 2\mu/\sigma$ obtain

$$\frac{d\alpha}{d\mu} = \frac{1}{\alpha\sigma}.$$

Let $v = y - \frac{1}{2}$. The derivative of K_v is

$$\frac{d}{dx} K_v(x) = -\frac{1}{2} (K_{v-1}(x) + K_{v+1}(x)).$$

Gradient w.r.t. μ

Differentiate ℓ term by term, remembering σ is constant and only $\alpha(\mu)$ and $\log \mu$ depend on μ :

$$\frac{\partial \ell}{\partial \mu} = \underbrace{\frac{1}{2} \cdot \frac{1}{\alpha} \cdot \frac{1}{\alpha \sigma}}_{\text{from } \frac{1}{2} \log(2\alpha/\pi)} + \underbrace{y \left(\frac{1}{\mu} - \frac{1}{\alpha} \cdot \frac{1}{\alpha \sigma} \right)}_{\text{from } y(\log \mu - \log(\alpha \sigma))} + \underbrace{\left(\frac{\partial}{\partial \alpha} \log K_v(\alpha) \right) \cdot \frac{1}{\alpha \sigma}}_{\text{from } \log K_v(\alpha)}.$$

Using $\frac{\partial}{\partial \alpha} \log K_v(\alpha) = K'_v(\alpha)/K_v(\alpha) = -\frac{1}{2} \{K_{v-1}(\alpha) + K_{v+1}(\alpha)\}/K_v(\alpha)$, this can be written compactly as

$$\boxed{\frac{\partial \ell}{\partial \mu} = \frac{1}{2\alpha^2 \sigma} + y \left(\frac{1}{\mu} - \frac{1}{\alpha^2 \sigma} \right) - \frac{1}{2\alpha \sigma} \frac{K_{v-1}(\alpha) + K_{v+1}(\alpha)}{K_v(\alpha)}, v = y - \frac{1}{2}.}$$

Gradient w.r.t. the linear predictor η

Because $\mu = \exp(\eta)$,

$$\frac{\partial \mu}{\partial \eta} = \mu \Rightarrow \boxed{\frac{\partial \ell}{\partial \eta} = \mu \frac{\partial \ell}{\partial \mu}}$$

and the per-observation gradient contribution for parameters θ is

$$\frac{\partial \ell}{\partial \theta} = \frac{\partial \ell}{\partial \eta} \frac{\partial \eta}{\partial \theta} = \left(\mu \frac{\partial \ell}{\partial \mu} \right) \frac{\partial f_{\theta}(\mathbf{x})}{\partial \theta},$$

since $\eta = f_{\theta}(\mathbf{x}) + \log(\text{Exposure})$ and the offset has zero derivative.

In mini-batch SGD/Nadam, these terms are averaged (or summed) over the batch.

Deviance (model assessment)

For the P-IG family, a model deviance analogous to GLMs compares the fitted model to the saturated model.

Let $\hat{\mu}_i$ denote the fitted mean for observation i .

The *per-observation* deviance contribution is

$$d_i = 2 [\log f(y_i \mid \mu_i = y_i, \sigma) - \log f(y_i \mid \mu_i = \hat{\mu}_i, \sigma)].$$

Averaging over a subset $\mathcal{A}$ gives the mean deviance:

$$\mathcal{L}_{\mathcal{A}} = \frac{2}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} [\log f(y_i \mid \mu_i = y_i, \sigma) - \log f(y_i \mid \mu_i = \hat{\mu}_i, \sigma)],$$

where $\mu_i = y_i$ corresponds to the saturated model, which achieves the maximal value of the log-likelihood.

This definition coincides with the Poisson deviance used in Section [7.2](#).

8. Conclusions

This report showed how data science and ethics can work together in insurance. It started with the main ethical ideas — privacy, fairness, transparency, human oversight, and solidarity — to make sure models are used responsibly and follow regulations.

For the travel insurance data, extreme and negative claim values were removed to clean the data. A Box–Cox transformation with $\hat{\lambda} \approx 0.02$ (almost a log transformation) helped reduce skewness and stabilise the data. Age bands were created to compare claims between different age groups, and males showed slightly higher average losses.

For the property insurance data, Principal Component Analysis (PCA) was used to reduce the number of variables while keeping most of the information. The first three components explained about 88.6% of the total variation. A regression model using these components showed a very small R^2 value, meaning that property features alone could not predict claim cost well. The model predicted a cost of about 178,000 for the new policy in Section [6.5](#).

For the motor insurance data, a Poisson deep neural network was trained using the tanh activation function and a log-exposure offset. The model included embeddings for vehicle brand and region to capture hidden patterns. Training and testing results were very similar (0.3117 vs 0.3069 deviance), showing the model worked well and did not overfit.

Overall, the report showed a complete process — from ethical thinking to advanced modelling — that combines traditional actuarial methods with modern machine learning. The results highlight that while classical models give clear understanding, neural networks can capture more complex patterns. Both are important for accurate, fair, and responsible insurance modelling.

9. References (Harvard Style)

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Appendix A — Full Package Details

The following R packages were used in this project:

Package	Primary Purpose
tidyverse	Data manipulation, piping, general workflows
dplyr	Data wrangling, filtering, summarizing
readxl	Import Excel data
ggplot2	Plotting & visualization
MASS	Box–Cox transformation
magrittr	Pipe operators
stats	PCA (princomp), linear modelling
keras3	Deep neural network modelling
reticulate	Python interface required by keras3
base	Core functionality
utils	Data import/export

Full Code

All R scripts used to produce the results in this report are provided in `appendix.txt`, including:

- Data import
- Pre-processing routines
- Box–Cox transformation
- PCA, biplot, PC regression
- Deep Poisson network with embeddings
- Model evaluation and deviance computations

Each script is fully commented to support comprehension and independent execution.